Test-Time Physical Awareness for Generated Humanoid Reference Motions

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Abstract

Modern humanoid motion trackers such as SONIC can follow rich whole-body reference trajectories in physics, but those trackers still need high-quality target motions. KIMODO makes reference authoring much easier by generating Unitree G1 kinematic motions from text prompts and constraints. This project studies the interface between those systems: given a generated G1 reference, can we detect physical risks, attempt a simple test-time repair, and then gate the result with a controller rollout?

I built a physical-awareness loop for generated reference motions. It evaluates generated references with MuJoCo inverse-dynamics/contact diagnostics [1] and a pretrained SONIC controller audit. The evaluator emits structured failure records such as joint-level torque exceedance, root-wrench demand, self-contact pairs, non-foot floor contacts, weak support, contact artifacts, and controller trackability failure. Those records are designed to drive either an LLM prompt-refinement pass or a direct reference-repair pass. The measured repair baseline in this release uses simple retiming/smoothing variants, and the same physical and SONIC gates decide whether a candidate is accepted, flagged, or rejected.

On the 100-prompt KIMODO benchmark, first-pass references pass the physical screen in 48/100 cases, complete the nominal 4-second SONIC rollout in 53/100 cases, and pass both gates in 29/100 cases. The deterministic repair snapshot improves physical pass from 48/100 to 53/100, SONIC no-fall from 53/100 to 56/100, mean SONIC tracking time from 2.855 s to 3.007 s, and mean RMSE from 0.156 to 0.142. It also has regressions. The contribution is not a solved text-to-robot motion generator; it is a concrete evaluation, repair, and acceptance-gate artifact for exposing generated-reference brittleness.

1 Problem

Humanoid motion control has recently improved because learned policies can be trained to track diverse reference motions under physics. SONIC is a strong example: it trains a Unitree G1 whole-body controller on over 100M frames / 700 hours of high-quality human motion [2]. At the input level, the tracker receives a target trajectory: root pose and joint targets over time. This follows a broader line of physics-based motion imitation and reusable skill learning, including DeepMimic [3], AMP [4], Perpetual Humanoid Control [5], ASE [6], and CALM [7].

The upstream question is where those references come from. KIMODO generates controllable kinematic human and robot motions from text prompts and constraints [8]. This makes it a natural source of candidate references for robot training and controller testing. However, a kinematic reference is not automatically physically executable. It may demand unrealistic torques, require impossible support, create self-collisions, drag contacts, or destabilize a learned controller. This reference-quality issue is also central to motion-retargeting work such as GMR, which studies how retargeting artifacts affect downstream humanoid tracking policies [9]. The text-to-motion side has also advanced through datasets and generators such as HumanML3D [10], T2M-GPT [11], MDM [12], MotionDiffuse [13], and MoMask [14]; this project asks what happens when such generated references are handed to a physics tracker rather than only rendered kinematically.

The project question is:

> Can a physical-awareness layer identify risky KIMODO-generated humanoid reference motions, and can those failure tags guide useful test-time repair before controller execution?

2 System

The final system is a test-time reference screening and repair loop. The loop is deliberately modular: the evaluator produces a structured description of what failed, and a repair policy decides what candidate to try next. In this release, the measured repair policy is deterministic retiming/smoothing. The same failure records also define an LLM prompt-refinement interface for regenerating future KIMODO candidates.

Stage	Input	Contribution	Output
Generate	text prompt	KIMODO G1 generation	root + joint trajectory
Evaluate	reference	MuJoCo + SONIC checks	risk flags and rollout metrics
Repair	failed candidate	retiming/smoothing or prompt refinement	repaired candidates
Repeat/gate	new candidate	rescore until budget ends	accepted or flagged clip

Failure tags are intentionally interpretable. High torque/root wrench signals abrupt motion; self-contact signals limb/body intersection risk; weak support signals a likely balance problem; and trackability failure means that the pretrained SONIC controller could not follow the reference for the rollout horizon. SONIC is used as a foundation-style tracking evaluator and acceptance gate, not as an optimizer that changes KIMODO.

2.1 Structured Failure Records

The evaluator does not only return a scalar risk. It returns records that are specific enough to explain the failure and to condition a repair rule. Table 1 shows the record fields used by the repair interface.

Table 1: Failure records emitted by the screen. Each record contains a metric tag plus local evidence that can be passed to a deterministic repair rule or an LLM prompt-refinement step.

Failure mode	Example record fields	Repair signal
Torque/root wrench	joint name, peak torque ratio, frame, root force	slow/smooth the offending limb
Self-contact	body pair, contact fraction, first frame	separate limbs/torso
Non-foot floor	body name, contact frames, task contact allowance	lift or justify non-foot support
Support proxy	support margin, low-support frame fraction	widen stance or reduce COM shift
Contact artifact	foot slip, penetration, impulse spike	clean foot placement
SONIC trackability	fall frame, track seconds, RMSE, root error	lower speed/amplitude

An illustrative serialized torque/root-wrench record looks like:

```
failure=torque_root_wrench; joint=right_shoulder_pitch;
peak_torque_limit_ratio=16.7; frame=84;
root_force_newton=6900; sonic_track_seconds=2.78.
```

For the LLM prompt-refinement path, this record is inserted into a constrained rewrite prompt:

Base instruction. Rewrite the humanoid motion prompt so it preserves the task intent but reduces the listed physical failure. Keep the motion on a Unitree G1 humanoid, avoid adding objects not in the original task, and produce one concise revised prompt.
Input. Original prompt, failure tag, structured metric record, and allowed repair vocabulary.
Output. A revised prompt such as "walk forward with slower, smoother arm swing and shorter steps while keeping both feet under the body."

This LLM interface is designed to turn simulator evidence into a generator-facing instruction. However, the aggregate repair numbers reported in this paper are from the deterministic retiming/smoothing baseline; prompt regeneration should be counted separately once a full regeneration run is available.

3 Benchmark and Metrics

The benchmark contains 100 humanoid robotics prompts grouped as locomotion and recovery (26), manipulation and loco-manipulation (26), floor/low posture (12), dance/expressive motion (12), athletic and terrain stress (16), and communication/safety gestures (8). The suite includes walking, shuffling, crawling, carrying, panel pressing, gestures, obstacle-style steps, forward rolls, cartwheel attempts, jumping, and low-posture motions.

For each generated G1 trajectory, the evaluator computes torque demand relative to actuator limits, root force/torque, velocity, acceleration, jerk, self-contact, non-foot floor contact, contact artifacts, support-region proxy violations, SONIC no-fall, survival time, and joint RMSE.

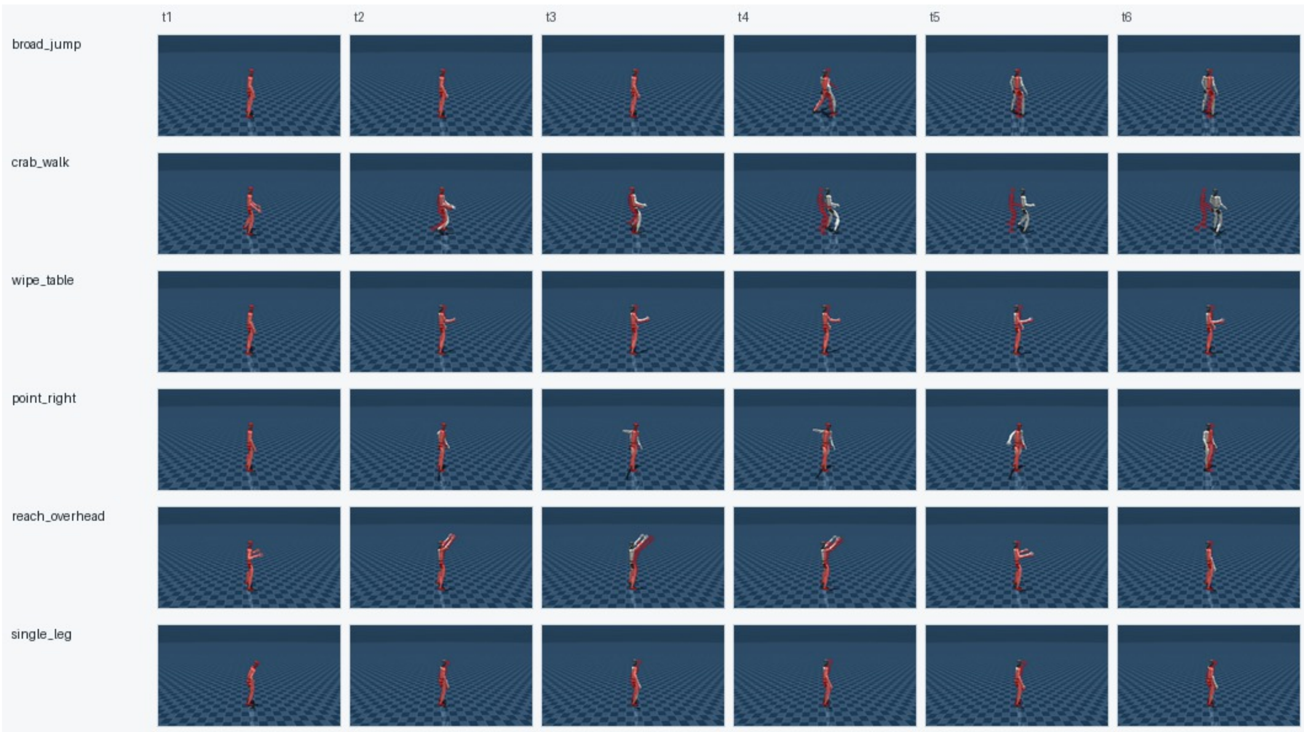

Figure 1: Six examples that pass the physical screen and complete the nominal SONIC rollout. Each row is one video sampled at six times. Red ghost is the reference target; solid G1 is the MuJoCo/SONIC rollout.

4 Results

4.1 First-Pass Audit

Table 2: First-pass 100-prompt KIMODO audit.

Set	Physical pass	SONIC no-fall	Both	Mean sec.	RMSE
All KIMODO refs	48/100	53/100	29/100	2.855	0.156
Physical-pass subset	48/48	29/48	29/48	3.293	0.170
Flagged subset	0/52	24/52	0/52	2.450	0.143

Only 29 of 48 physical-pass references also complete SONIC. Meanwhile, 24 of 52 flagged references still complete SONIC. This disagreement is a central result: reference-level physical checks and controller-level trackability are both needed.

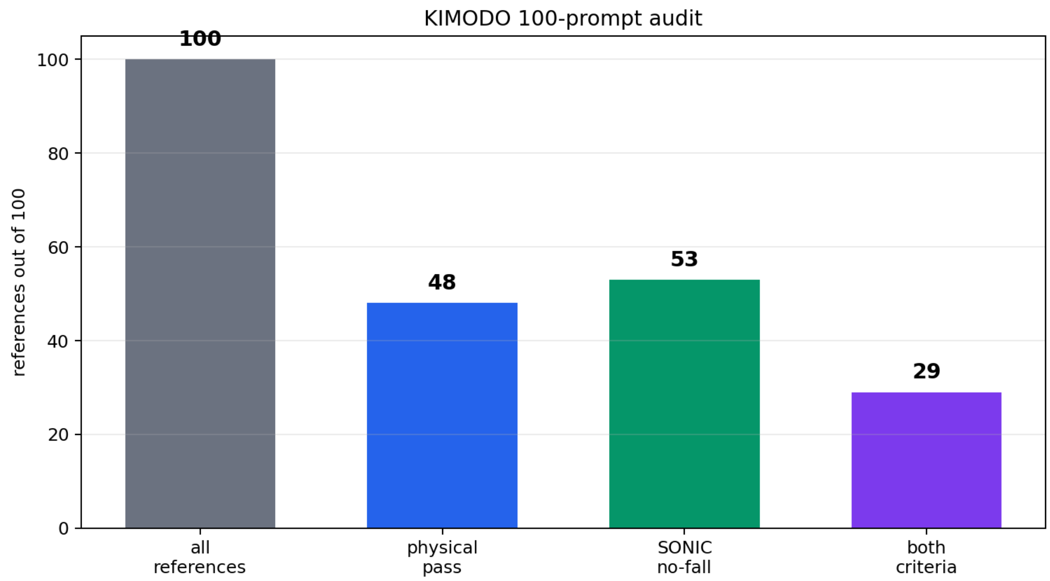

Figure 2: First-pass KIMODO audit over 100 prompts. The physical screen and SONIC controller rollout are related but not equivalent.

Table 3: First-pass non-exclusive failure tags.

Failure tag	Count
Physical screen fail	52
SONIC fall	47
Torque limit >1x	66
High root force >5 kN	28
Self-contact >8%	34
Non-foot floor contact	11
Low foot support	7
Contact artifact >0.45	15
Floor penetration >8 cm	10

4.2 Failure Tags

The representative failure videos in the deck are 'forward_roll' for torque/root wrench, 'turn_in_place' for self-contact, 'knee_slide' for non-foot floor contact, 'step_over_right' for support proxy, 'carry_box' for controller trackability, and 'happy_dance' for contact artifact.

4.3 Deterministic Repair Snapshot

Here, "critic reject status" means the candidate is not accepted by the current gate and should trigger another attempt; it is not counted as evidence that a full KIMODO prompt-regeneration run was performed.

The repair pass produced seven SONIC rescues and four regressions. The main repair videos show representative improvements: 'forward_walk' becomes slower and smoother after a trackability flag; 'wave' uses planted feet and smooth arm motion after torque/tracking flags; 'split_squat_jump' uses a smaller jump and soft landing after support/torque flags; 'happy_dance' improves contact placement; and 'backward_recovery' uses a smaller lean and step back. The deck also includes an illustrative 'pushup_pose' height-clearance demo; the table above is the measured evidence.

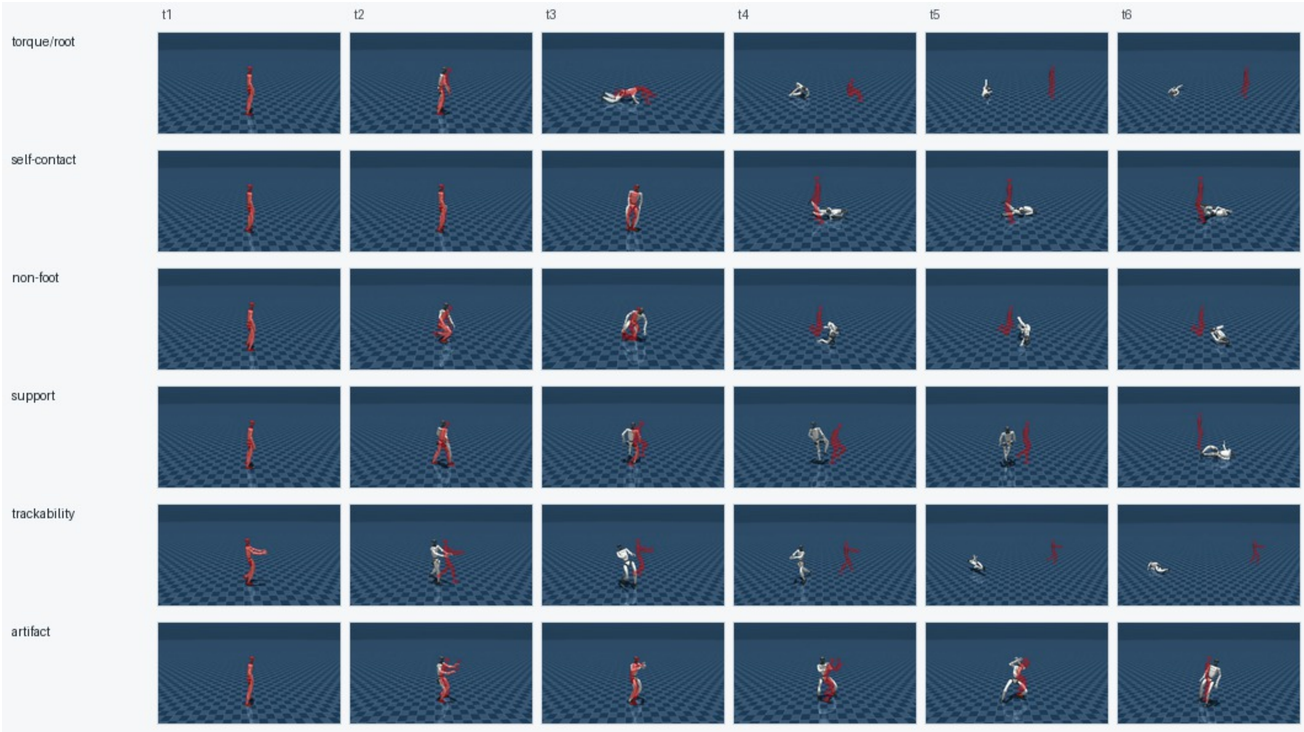

Figure 3: Representative generated-reference failures used in the slide deck. Rows correspond to torque/root-wrench demand, self-contact, non-foot floor contact, support proxy, controller trackability, and contact artifact. Each row shows six frames from the corresponding rollout video.

5 Interpretation

The results support the hypothesis in a bounded way. Generated references have measurable physical failure modes; failure tags are useful for curation and simple repair; and SONIC is stricter than kinematic replay. Human visual review is used as a sanity check for presentation examples, not as the source of metric labels.

The boundary slide shows remaining failures: 'forward_roll' for torque/root wrench stress, 'carry_box' for controller trackability, and 'cartwheel_attempt' for acrobatics. The current method does not solve arbitrary text-to-robot motion, robust acrobatics/low-posture tracking, or automatic repair for every invalid reference. Stronger results likely require fine-tuning or training a motion-level refiner, not only retiming and smoothing.

6 Limitations and Future Work

Deterministic retiming/smoothing repair helps some clips but is not a general motion generator. The physical screen is heuristic and simulator-dependent. SONIC rollout is a simulation diagnostic, not real hardware validation. Some motions need semantic context: floor contact can be correct for crawling and wrong for walking. Prompt fidelity is checked qualitatively; there is no full human-study or vision-language benchmark.

Future work should train a risk-conditioned refiner that learns from failure tags and SONIC rollouts, use short controller rollouts or a learned surrogate during candidate selection, and improve semantic evaluation so physical stability does not hide prompt mismatch.

Table 4: Measured retiming/smoothing repair snapshot.

Metric	Original	Repaired
Physical pass	48/100	53/100
Critic accept	47/100	54/100
Critic reject status	18/100	16/100
SONIC 4 s no-fall	53/100	56/100
Mean SONIC seconds	2.855	3.007
Mean RMSE	0.156	0.142
Mean risk	41.548	37.916
Contact artifact	0.264	0.247

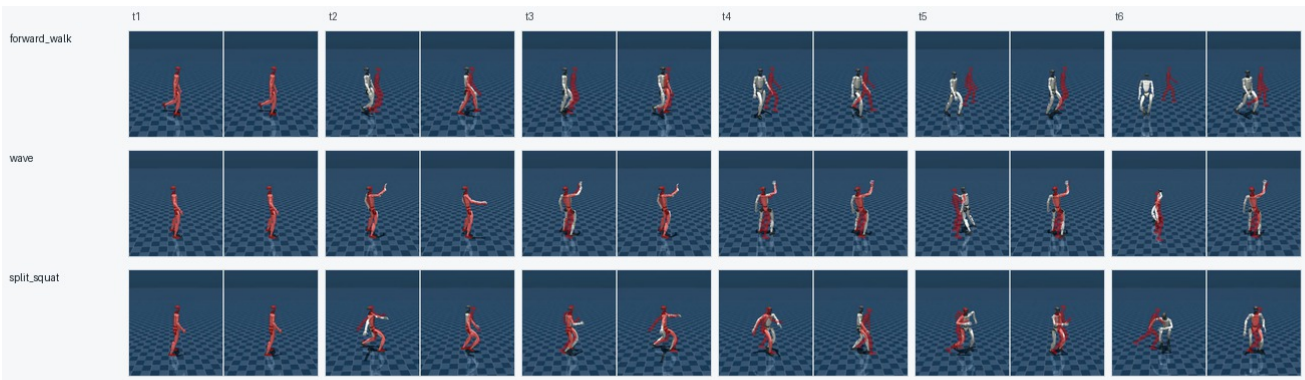

Figure 4: Before/after repair videos sampled across time. Each panel is a paired video: original KIMODO on the left and the selected repaired candidate on the right. The three rows show forward walking, waving, and split-squat jump repair examples from the deck.

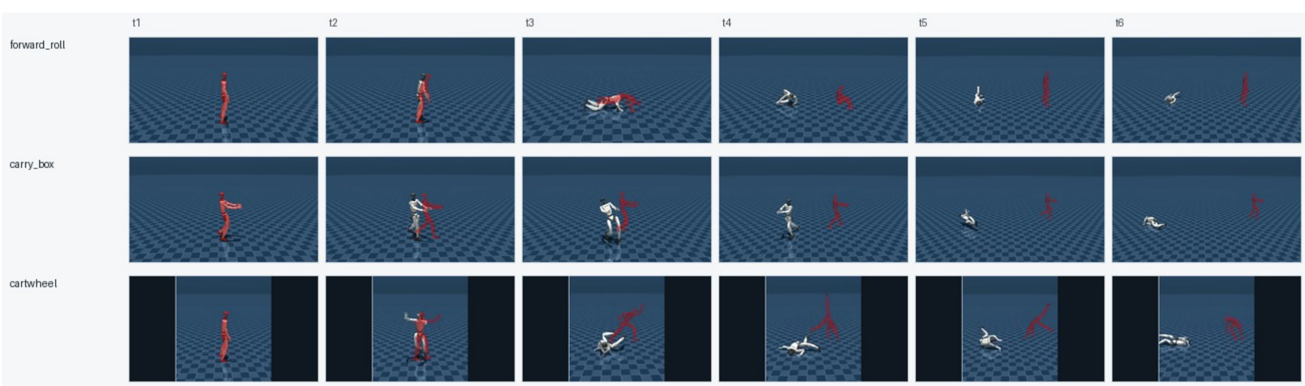

Figure 5: Boundary cases that remain hard after the current screen/repair loop: forward roll, carry-box tracking, and cartwheel-style acrobatics. These examples are included to make the negative result visible, not to claim success.

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